

Ability of a high-total antioxidant capacity diet to increase stool weight and bowel antioxidant status in human subjects.

There is limited knowledge about the possible effect of unabsorbed dietary antioxidants that reach the large intestine on bowel habits. The aim of the present study was to investigate whether a dietary recommendation directed to increase diet total antioxidant capacity (TAC) is able to affect gut function in human subjects. In this cross-over intervention, nineteen subjects followed a high-TAC (HT) and a low-TAC (LT) diet for 2 weeks, which were comparable for energy, macronutrient, total dietary fibre and alcohol contents. At the end of each intervention period, the 48 h stool output was recorded. In the faecal samples obtained from a subset of nine subjects, moisture, pH, ammonia content, Lactobacillus and Bifidobacterium counts, faecal water antioxidants and genotoxicity were measured. A 3 d weighed food record was used to assess the diet composition during HT and LT diet intake. Significant increases in the intake of TAC, vitamins E and C and phenolic compounds were observed during the HT diet intake. The higher intake of antioxidants led to increased 48 h stool output (324 (SD 38) g in HT v. 218 (SD 22) g in LT), and to higher TAC and total phenolic concentrations in faecal water. No significant variation in the other measured parameters was observed between the diets. In conclusion, a diet selected to raise the intake of dietary antioxidants is able to increase stool bulk and antioxidant content of faeces.